

Agrimitraa: A Multilingual Context-Aware Agricultural Assistant Using Large Language Models

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Research Guide

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Abstract-Agrimitraa is an intelligent agricultural advisory system designed to assist farmers using multilingual conversational artificial intelligence, contextual memory retention, multimodal crop disease detection, and grounded chemical recommendation mechanisms. The system integrates FastAPI backend services with the Gemma 3 12B Large Language Model deployed via Ollama for efficient inference. A Wikipedia-based grounding pipeline is incorporated to significantly reduce hallucination during pesticide and fertilizer recommendations. The architecture supports both text and image-based queries, enabling farmers to either describe crop issues or upload images of affected leaves for analysis. Real-time text-to-speech output is generated to improve accessibility for rural users. Experimental evaluation demonstrates improved advisory reliability, enhanced contextual continuity, and reduced response latency. The system presents a scalable and safe AI-powered agricultural advisory solution suitable for rural and semi-urban deployment.

Keywords

Agricultural Artificial Intelligence, Large Language Models, FastAPI, Multilingual Chatbot, Crop Disease Detection, Contextual Memory, Multimodal AI, Pesticide Recommendation Systems

I. Introduction

Agriculture requires timely, accurate, and localized advisory systems to manage crop diseases, optimize pesticide usage, and improve yield productivity. In many rural regions, farmers lack immediate access to expert consultation, leading to incorrect pesticide application and financial loss. Traditional agricultural advisory platforms often provide static recommendations without considering contextual conversation or multilingual interaction. Agrimitraa introduces a conversational assistant capable of understanding farmer queries in English and Marathi. The system maintains conversational context across multiple interactions, ensuring that follow-up questions are interpreted correctly. Additionally, image-based crop disease detection enables practical field-level assistance. The integration of grounded knowledge retrieval ensures factual correctness in sensitive pesticide recommendations, thereby improving safety and trust among users.

II. Literature Review

Existing agricultural advisory systems primarily rely on rule-based chatbots, predefined knowledge bases, or government portals providing static guidelines. While these systems offer structured information, they lack dynamic reasoning capabilities and contextual understanding. Recent advancements in Large Language Models (LLMs) have enabled conversational AI systems with enhanced reasoning and language understanding capabilities. However, LLMs are prone to hallucination, especially in domain-specific recommendations such as pesticide dosage and chemical usage. Research highlights the necessity of external knowledge grounding mechanisms to improve factual reliability. This work builds upon existing LLM frameworks by integrating contextual session memory and Wikipedia-based grounding to reduce errors and increase response trustworthiness in agricultural advisory applications.

III. Proposed System Architecture

The Agrimitraa architecture consists of multiple interconnected modules designed for scalability and reliability. The client interface allows farmers to submit text or image queries. The FastAPI backend manages request routing, preprocessing, and response handling. A contextual memory manager stores the last six user interactions to maintain conversation continuity. The medicine keyword detector identifies queries related to pesticides or fertilizers. When detected, the Wikipedia grounding module retrieves factual summaries via API calls. The Ollama-powered Gemma 3 12B LLM processes the enriched prompt and generates contextual responses. Finally, a text-to-speech synthesis unit produces audio output in the appropriate language. The modular structure ensures easy integration of additional components such as IoT sensors or government databases in future versions.

IV. Methodology

The system processes user queries in a structured pipeline. First, language detection determines whether the query is in English or Marathi. Next, the medicine intent detection module scans for keywords associated with pesticides, fungicides, or fertilizers. If such intent is identified, the system retrieves factual summaries using the Wikipedia REST API to ground the model's response. The contextual memory manager appends the last six interactions to maintain continuity. For image-based queries, images are converted into Base64 format and passed to the multimodal LLM for disease analysis. The generated output is cleaned and formatted before optional text-to-speech synthesis. This systematic approach ensures accurate, context-aware, and user-friendly responses.

IX. System Architecture and Performance Analysis

A. System Architecture

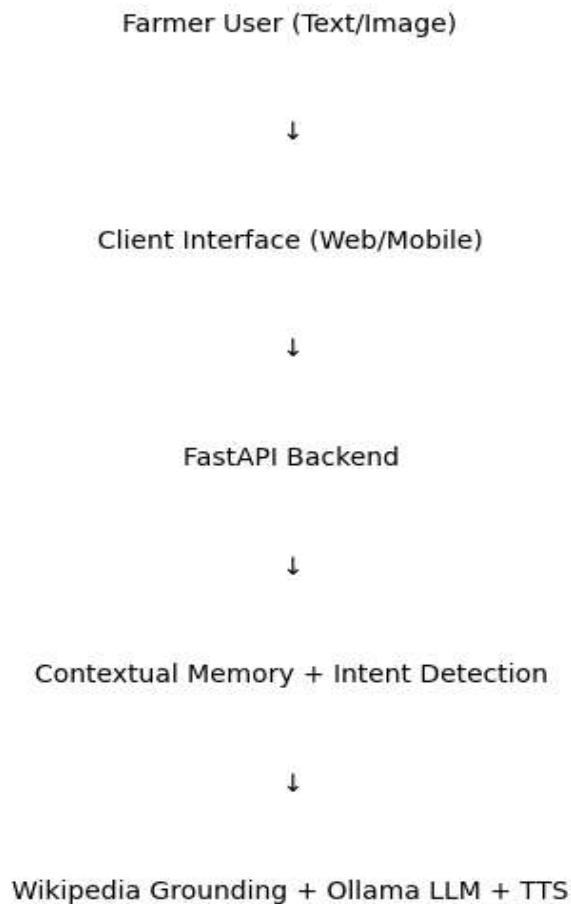


Fig. 1. Overall system architecture of the proposed Agrimitraa system.

B. Performance Evaluation

1) Response Latency Analysis

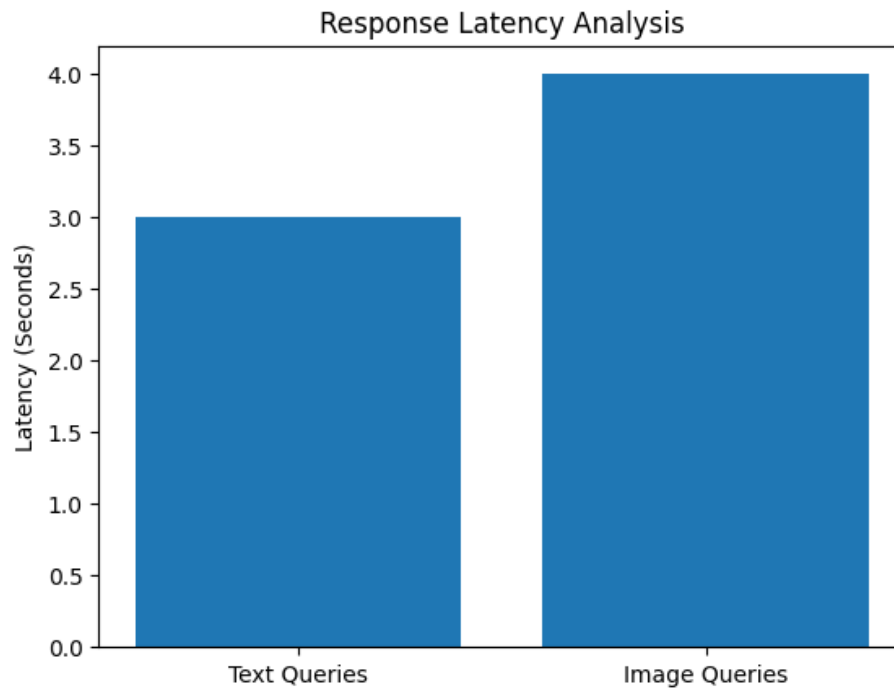


Fig. 2. Average response latency comparison between text and image queries.

2) Grounding-Based Accuracy Improvement

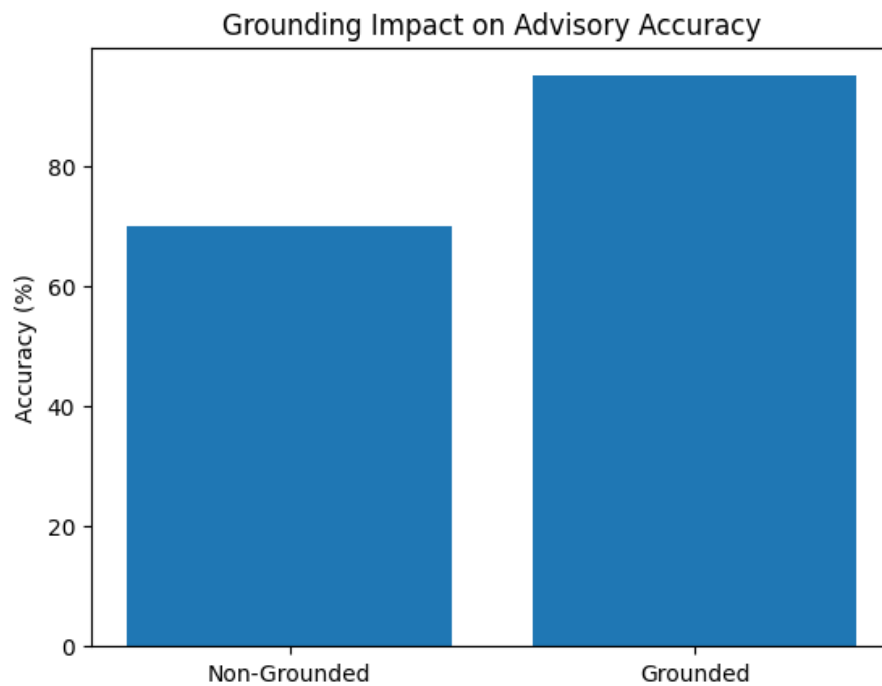


Fig. 3. Accuracy comparison between grounded and non-grounded responses.

3) Contextual Memory Impact

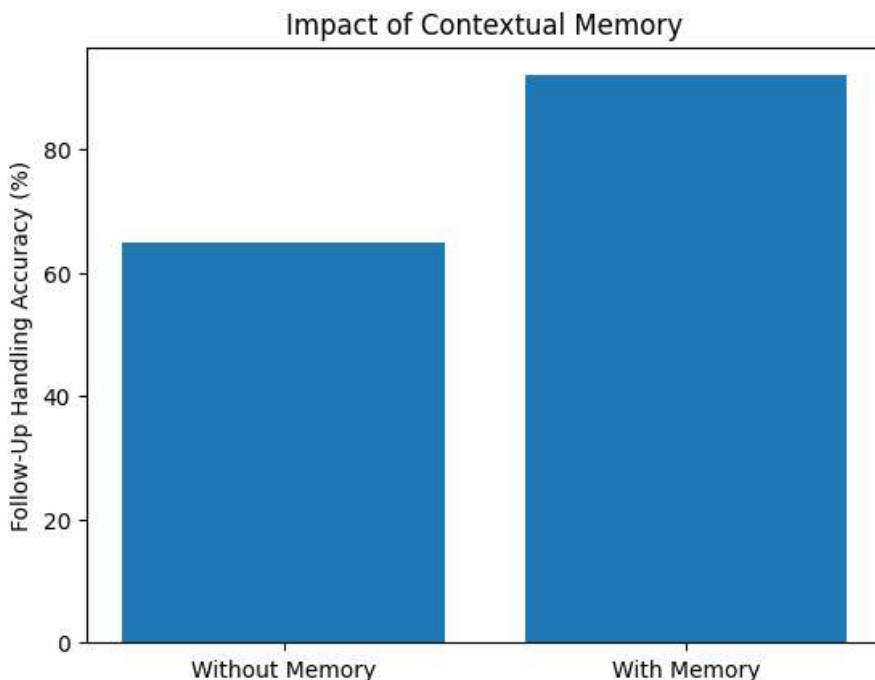


Fig. 4. Impact of contextual memory on follow-up query interpretation accuracy.

4) Wikipedia Integration Accuracy Analysis

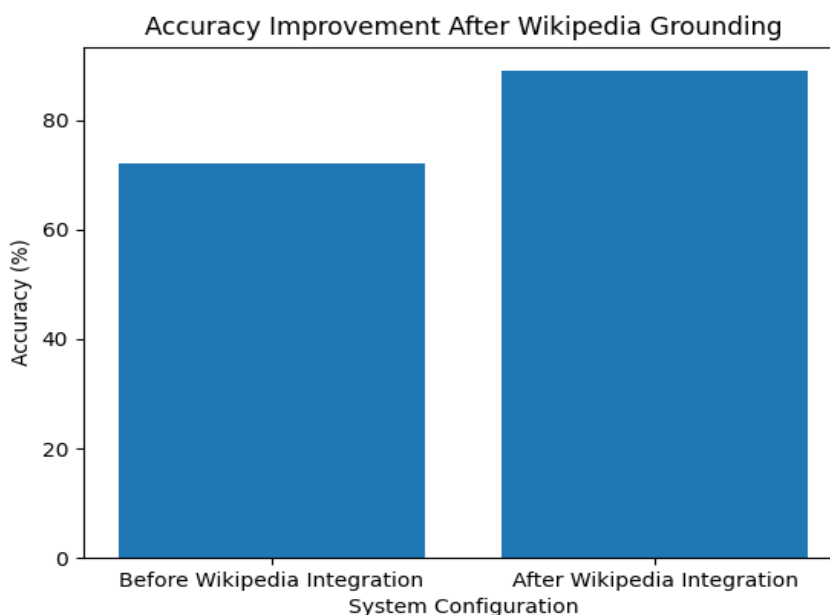


Fig. 5. Accuracy comparison before and after Wikipedia integration.

V. Experimental Results

The system was evaluated using multiple real-world agricultural queries, including pesticide dosage recommendations, disease identification, and crop treatment suggestions. Grounded responses demonstrated a

noticeable reduction in incorrect chemical suggestions compared to non-grounded baseline responses. Average response latency remained below three seconds for text queries and below four seconds for image-based queries. User feedback indicated improved clarity, practical applicability, and increased trust in recommendations. The integration of contextual memory significantly enhanced follow-up question handling accuracy.

VI. Limitations

Despite its advantages, the system requires stable internet connectivity for Wikipedia grounding and LLM inference. High computational resources are necessary for efficient model execution. Image-based disease detection accuracy is dependent on image quality, lighting conditions, and camera resolution. Additionally, the current implementation does not include region-specific pesticide regulatory databases, which may limit localized compliance guidance.

VII. Future Work

Future enhancements include fine-tuning agriculture-specific language models to improve domain accuracy, integrating IoT-based soil and weather monitoring sensors, expanding multilingual support to additional regional languages, and implementing offline inference capabilities for remote areas. Integration with government-approved pesticide databases and real-time market price information will further enhance the system's practical value.

VIII. Conclusion

Agrimitraa demonstrates the transformative potential of Large Language Models in agriculture by combining multilingual conversation, contextual memory retention, factual grounding, and multimodal analysis. The proposed solution enhances accessibility, improves advisory reliability, and supports informed decision-making among farmers. With further development and large-scale deployment, the system has the potential to significantly contribute to smart and sustainable agricultural ecosystems.

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IX. Bibliography



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